

2.6.30

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

If the area of a triangle is 35 square units with vertices $(2, 6)$, $(5, -4)$ and $(k, 4)$, then find k .

Solution:

$$\vec{A} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}, \quad \vec{B} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}, \quad \vec{C} = \begin{pmatrix} k \\ 4 \end{pmatrix}. \quad (0.1)$$

$$\vec{B} - \vec{A} = \begin{pmatrix} 3 \\ -10 \end{pmatrix}, \quad (0.2)$$

$$\vec{C} - \vec{A} = \begin{pmatrix} k-2 \\ -2 \end{pmatrix}. \quad (0.3)$$

$$\text{Area} = \frac{1}{2} |(\vec{B} - \vec{A}) \times (\vec{C} - \vec{A})|.$$

$$35 = \frac{1}{2} \left| \begin{pmatrix} 3 \\ -10 \end{pmatrix} \times \begin{pmatrix} k-2 \\ -2 \end{pmatrix} \right|.$$

$$35 = \frac{1}{2} |3(-2) - (-10)(k-2)| \quad (0.4)$$

$$= \frac{1}{2} |-6 + 10k - 20| \quad (0.5)$$

$$= \frac{1}{2} |10k - 26|. \quad (0.6)$$

Hence $|10k - 26| = 70$.

Solve the two linear cases:

$$10k - 26 = 70 \quad \Rightarrow \quad k = \frac{48}{5}, \quad (0.7)$$

$$10k - 26 = -70 \quad \Rightarrow \quad k = -\frac{22}{5}. \quad (0.8)$$

Therefore the possible values of k are $k = \frac{48}{5}$ or $k = -\frac{22}{5}$.

Triangles with vertices $(2,6)$, $(5,-4)$, and $(k,4)$ for $k=-4.4$ and $k=9.6$

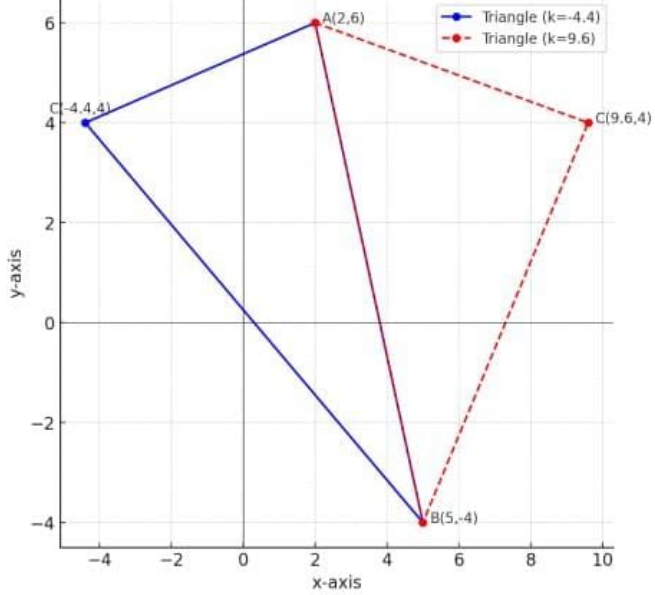


Fig. 0.1